Asssignment: Machine Learning

MAKINI CALEB

2026-10-15

SETTING A WORKING DIRECTORY

```
setwd("C:/Users/ADMIN/OneDrive/Desktop/caleb makini")
```

load libraries

```
library(tidyverse)
library(dplyr)
library(ggplot2)
library(ggthemes)
library(psych)
library(sjmisc)
library(kableExtra)
library(knitr)
library(tidyr)
library(reshape2)
library(formatR)
library(corrplot)
```

DATASET 1: LOAN PREDICTION (Classification)

Target variable: loan Approved

Algorithm: SVM Classification

section A: Data Cleaning

Q1: Load dataset

```
data <- read.csv("Loan_Approved data.csv")
head(data)
```

```
  Applicant_ID Age Income Credit_Score Loan_Amount Employment_Years
1            1  25  35000          580       15000                2
2            2  34  62000          720       20000                8
3            3  45  85000          780       30000               15
4            4  29  28000          520       10000                1
5            5  52 120000          810       50000               20
6            6  23  22000          490        8000                0
  Loan_Approved
1             0
2             1
3             1
4             0
5             1
6             0
```

report

```
cat("NO of Rows:", nrow(data), "\n No of columns:", ncol(data))
```

```
NO of Rows: 592
 No of columns: 7
```

- The data set has 592 rows and 7 columns

Check the structure of the data set

```
str(data)
```

```
'data.frame':   592 obs. of  7 variables:
 $ Applicant_ID    : int  1 2 3 4 5 6 7 8 9 10 ...
 $ Age             : int  25 34 45 29 52 23 38 41 27 36 ...
 $ Income          : int  35000 62000 85000 28000 120000 22000 55000 7
1000 31000 48000 ...
 $ Credit_Score    : int  580 720 780 520 810 490 680 750 560 640 ...
 $ Loan_Amount     : int  15000 20000 30000 10000 50000 8000 25000 350
00 12000 18000 ...
 $ Employment_Years: int  2 8 15 1 20 0 10 12 3 7 ...
 $ Loan_Approved   : int  0 1 1 0 1 0 1 1 0 1 ...
```

columns that are incorrectly typed

- Loan_approved from integer to factor
- conversion to factor

```
data$Loan_Approved <- as.factor(data$Loan_Approved)
```

confirm for changes

```
str(data)
```

```
'data.frame':   592 obs. of  7 variables:
 $ Applicant_ID    : int  1 2 3 4 5 6 7 8 9 10 ...
 $ Age             : int  25 34 45 29 52 23 38 41 27 36 ...
 $ Income          : int  35000 62000 85000 28000 120000 22000 55000 7
1000 31000 48000 ...
 $ Credit_Score    : int  580 720 780 520 810 490 680 750 560 640 ...
 $ Loan_Amount     : int  15000 20000 30000 10000 50000 8000 25000 350
00 12000 18000 ...
 $ Employment_Years: int  2 8 15 1 20 0 10 12 3 7 ...
 $ Loan_Approved   : Factor w/ 2 levels "0","1": 1 2 2 1 2 1 2 2 1 2
...
```

Q3: Check for duplicates

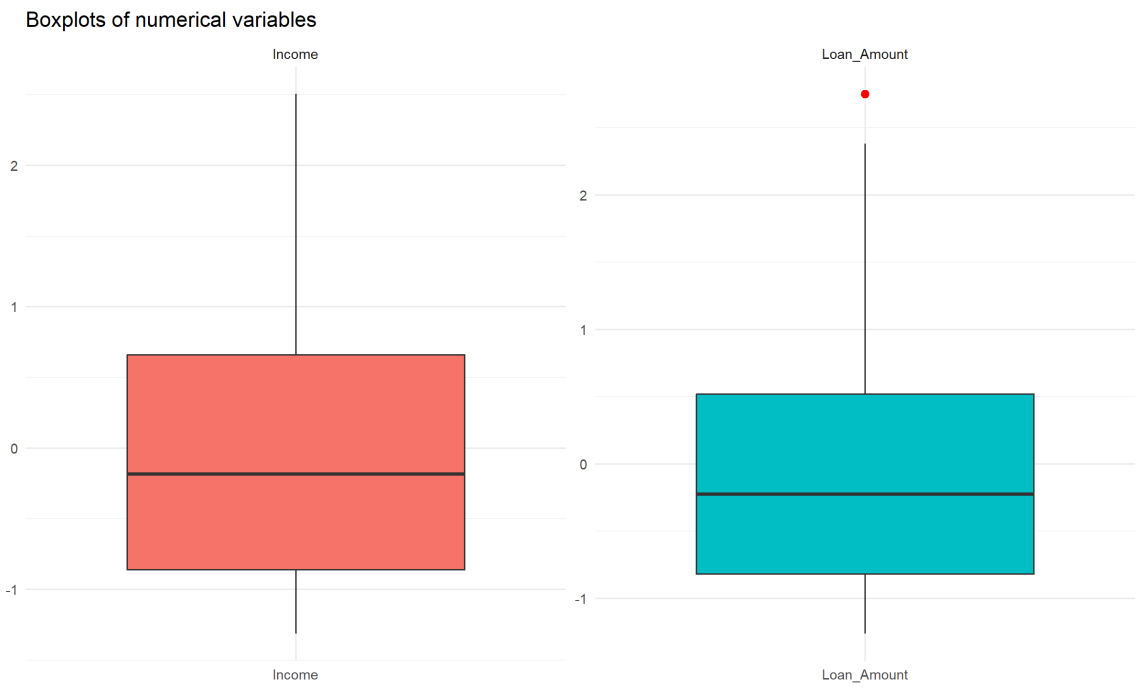
```
n_dups <- sum(duplicated(data))
cat(if (n_dups == 0) "No duplicates found.\n" else paste(n_dups, "dupl
icate row(s) found.\n"))
```

```
No duplicates found.
```

Q4: Detect and Identify outliers

select numeric variables

```
c_var <- c("Income", "Loan_Amount")
data[c_var] |>
    scale() |>
    as.data.frame() |>
    pivot_longer(everything(), names_to = "variable", values_to = "val
ue") |>
    ggplot(aes(x = variable, y = value, fill = variable)) + geom_boxpl
ot(outlier.colour = "red",
    outlier.size = 2) + facet_wrap(~variable, scales = "free") + theme
_minimal() +
    theme(legend.position = "none") + labs(title = "Boxplots of numeri
cal variables",
    x = NULL, y = NULL)
```



detection using 1.5 IQR

```
detect_outliers_IQR <- function(x) {
    Q1 <- quantile(x, 0.25, na.rm = TRUE)
    Q3 <- quantile(x, 0.75, na.rm = TRUE)
    IQR <- Q3 - Q1
    lower <- Q1 - 1.5 * IQR
    upper <- Q3 + 1.5 * IQR
    which(x < lower | x > upper)
}
```

confirmation of outlier detection

count outliers

```
outlier_count <- sapply(data[c_var], function(x) length(detect_outlier
s_IQR(x)))
outs <- outlier_count[outlier_count > 0]
if (length(outs) == 0) {
    cat("No outliers detected in any numerical variable.\n")
} else {
    cat("Outliers detected:\n")
    print(outs)
}
```

```
Outliers detected:
Loan_Amount
         12
```

get mode function

```
get_mode <- function(x) {
    ux <- unique(na.omit(x))
    ux[which.max(tabulate(match(x, ux)))]
}
cat("mode=", get_mode(data$Loan_Amount), "\n")
```

```
mode= 15000
```

replace with the mode (iterate until stable)

```
max_iter <- 10
converged <- FALSE
for (i in seq_len(max_iter)) {
    n_replaced <- 0
    for (var in c_var) {
        idx <- detect_outliers_IQR(data[[var]])
        if (length(idx) > 0) {
            mode_val <- get_mode(data[[var]])
            data[[var]][idx] <- mode_val
            n_replaced <- n_replaced + length(idx)
        }
    }
    cat(sprintf("Pass %d: replaced %d value(s)\n", i, n_replaced))
    if (n_replaced == 0) {
        converged <- TRUE
        break
    }
}
```

```
Pass 1: replaced 12 value(s)
Pass 2: replaced 12 value(s)
Pass 3: replaced 0 value(s)
```

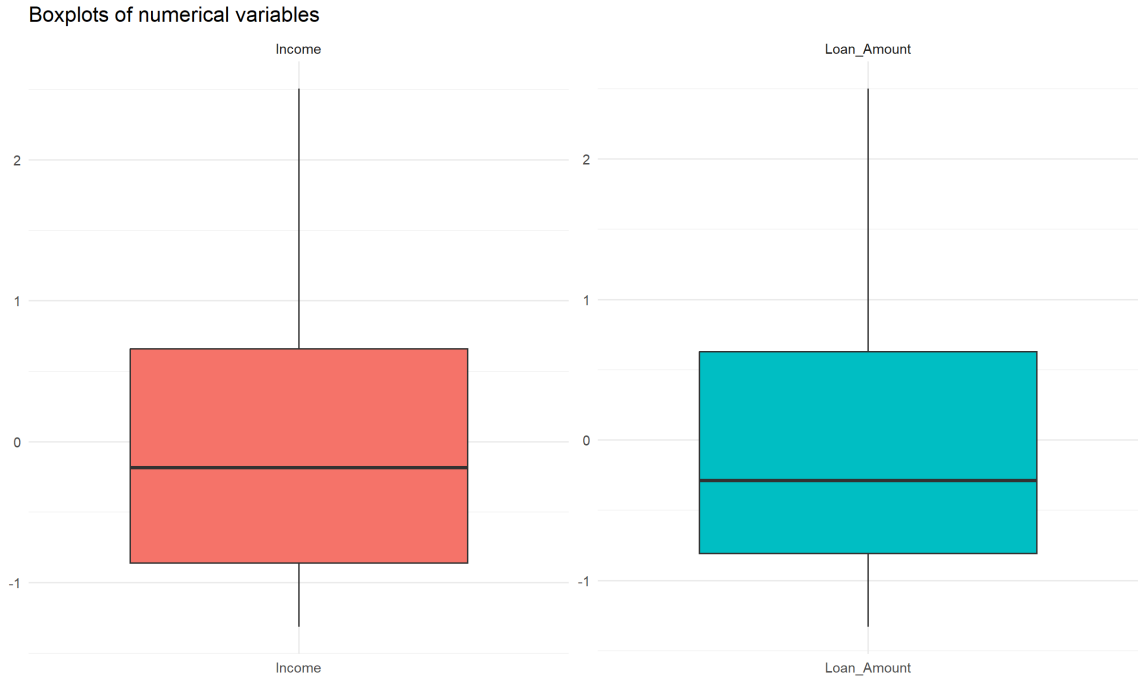
```
if (!converged) warning("Outlier replacement did not converge in ", ma
x_iter, " passes.")
```

re_count outliers per columns

```
outlier_count_after <- sapply(data[c_var], function(x) length(detect_o
utliers_IQR(x)))
outliers_check <- data.frame(before_treatment = outlier_count, after_t
reatment = outlier_count_after)
outliers_check
```

```
            before_treatment after_treatment
Income                     0               0
Loan_Amount               12               0
```

```
data[c_var] |>
    scale() |>
    as.data.frame() |>
    pivot_longer(everything(), names_to = "variable", values_to = "val
ue") |>
    ggplot(aes(x = variable, y = value, fill = variable)) + geom_boxpl
ot(outlier.colour = "red",
    outlier.size = 2) + facet_wrap(~variable, scales = "free") + theme
_minimal() +
    theme(legend.position = "none") + labs(title = "Boxplots of numeri
cal variables",
    x = NULL, y = NULL)
```



print a clear “none” message

```
total <- sum(outlier_count_after)
cat(if (total == 0) "No outliers remain.\n" else sprintf("\n%d outlier
(s) still present.\n",
    total))
```

```
No outliers remain.
```

```
for (var in c_var) {
    n <- length(detect_outliers_IQR(data[[var]]))
    cat(sprintf("%-18s : %s\n", var, if (n == 0)
        "clean"
    else paste(n, "outlier(s) remaining")))
}
```

```
Income             : clean
Loan_Amount        : clean
```

```
stopifnot(all(sapply(data[c_var], function(x) length(detect_outliers_I
QR(x))) ==
    0))
cat("Confirmed: no outliers detected in any column of c_var.\n")
```

```
Confirmed: no outliers detected in any column of c_var.
```

section B: MIssing variables

Q5: Count number of missing variables in each column

```
na_counts <- colSums(is.na(data))
miss <- na_counts[na_counts > 0]
if (length(miss) == 0) {
    cat("No missing values in any column.\n")
} else {
    cat("Columns with missing values:\n")
    print(miss)
}
```

```
No missing values in any column.
```

Q5 successful checked and no missing value hence we skip

- Q6. Impute missing values in `Income` and `Loan_Amount` using the median. Justify why median is preferred over mean for these columns.
- Q7. Would you impute missing values in the target variable `Loan_Approved` ? Justify your answer.
- Q8. Verify that no missing values remain after treatment.

section c: Descriptive statistics

compute mean, median, sd, minimum and maximumn of all numerical variables

select numerical variables

```
num_var <- c("Age", "Income", "Credit_Score", "Loan_Amount", "Employme
nt_Years")
```

confirmation of outlier detection

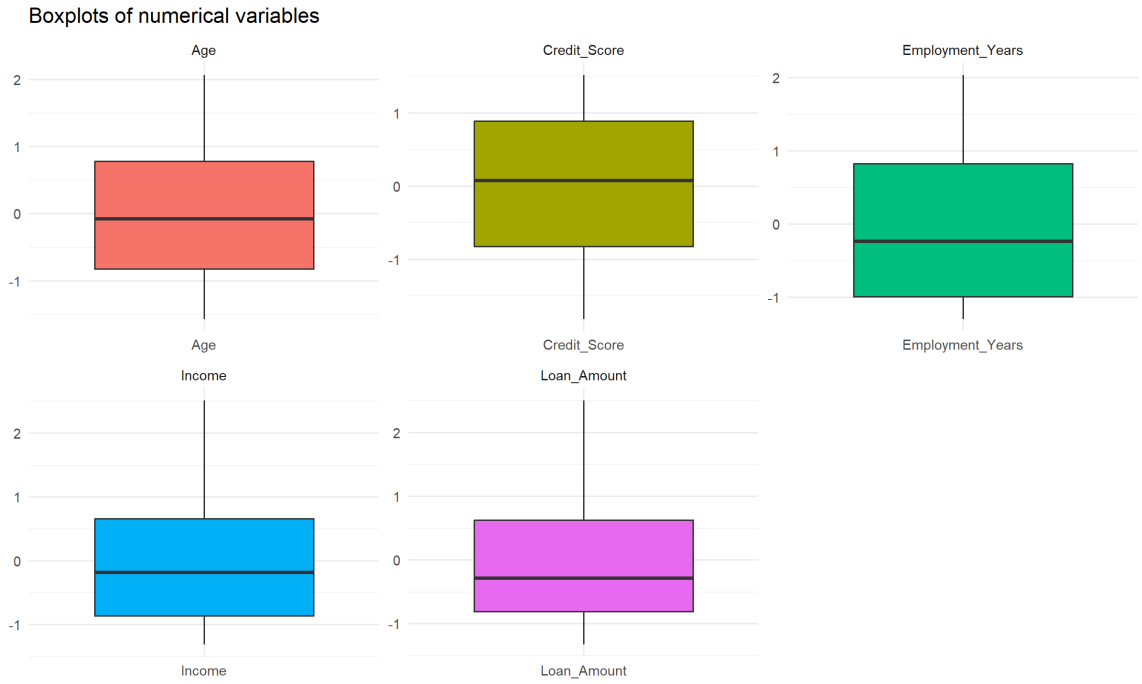
count outliers

```
outlier_count <- sapply(data[num_var], function(x) length(detect_outli
ers_IQR(x)))
outs <- outlier_count[outlier_count > 0]
if (length(outs) == 0) {
    cat("No outliers detected in any numerical variable.\n")
} else {
    cat("Outliers detected:\n")
    print(outs)
}
```

```
No outliers detected in any numerical variable.
```

count outliers per columns

```
data[num_var] |>
    scale() |>
    as.data.frame() |>
    pivot_longer(everything(), names_to = "variable", values_to = "val
ue") |>
    ggplot(aes(x = variable, y = value, fill = variable)) + geom_boxpl
ot(outlier.colour = "red",
    outlier.size = 2) + facet_wrap(~variable, scales = "free") + theme
_minimal() +
    theme(legend.position = "none") + labs(title = "Boxplots of numeri
cal variables",
    x = NULL, y = NULL)
```




```
summary_stats <- function(x) {
    c(mean = mean(x, na.rm = TRUE), sd = sd(x, na.rm = TRUE), min = mi
n(x, na.rm = TRUE),
        max = max(x, na.rm = TRUE))
}
stats <- sapply(data[num_var], summary_stats)
stats_df <- data.frame(variable = colnames(stats), t(stats), row.names
= NULL)
stats_df
```

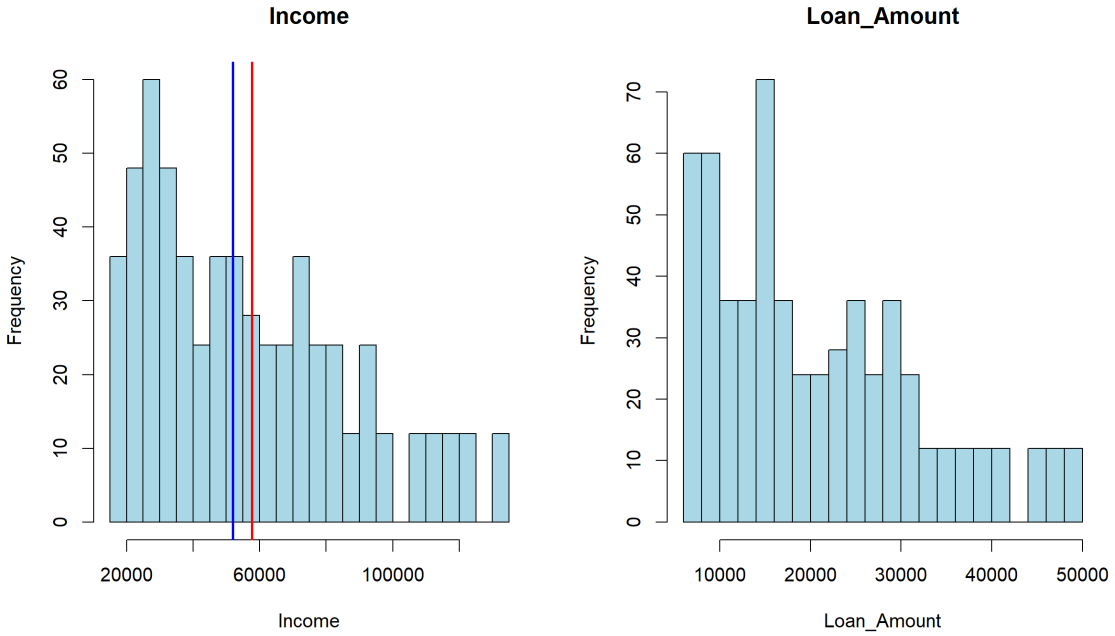
```
          variable         mean           sd   min    max
1              Age    35.682432     9.358465    21     55
2           Income 57641.891892 30869.631474 17000 135000
3     Credit_Score   651.655405   110.941805   450    820
4      Loan_Amount 21294.594595 11481.604702  6000  50000
5 Employment_Years     8.554054     6.601321     0     22
```

Q10: compute skewness

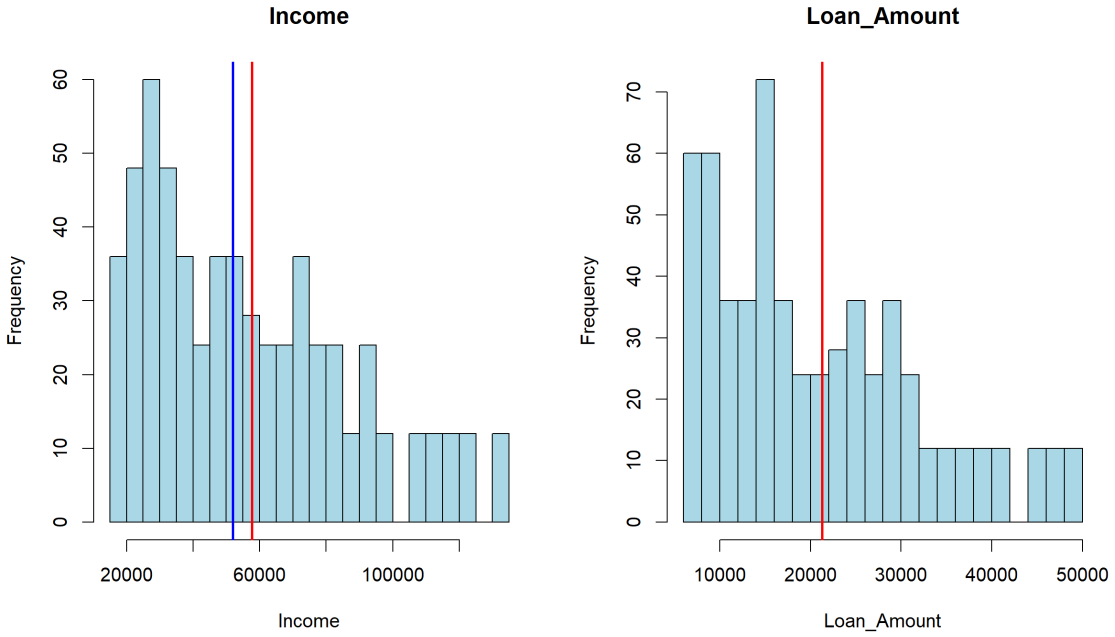
```
library(e1071)
sk <- sapply(data[num_var], skewness)
skew_df <- data.frame(Variable = names(sk), Skewness = round(as.numeri
c(sk), 3),
    Direction = ifelse(sk > 0, "Right-skewed (positive)", ifelse(sk <
0, "Left-skewed (negative)",
        "Symmetric")), Magnitude = ifelse(abs(sk) < 0.5, "Low / roughl
y symmetric",
        ifelse(abs(sk) < 1, "Moderate", "High")), row.names = NULL)
skew_df
```

```
          Variable Skewness               Direction               Magn
itude
1              Age    0.279 Right-skewed (positive) Low / roughly symm
etric
2           Income    0.713 Right-skewed (positive)               Mod
erate
3     Credit_Score   -0.148  Left-skewed (negative) Low / roughly symm
etric
4      Loan_Amount    0.749 Right-skewed (positive)               Mod
erate
5 Employment_Years    0.422 Right-skewed (positive) Low / roughly symm
etric
```

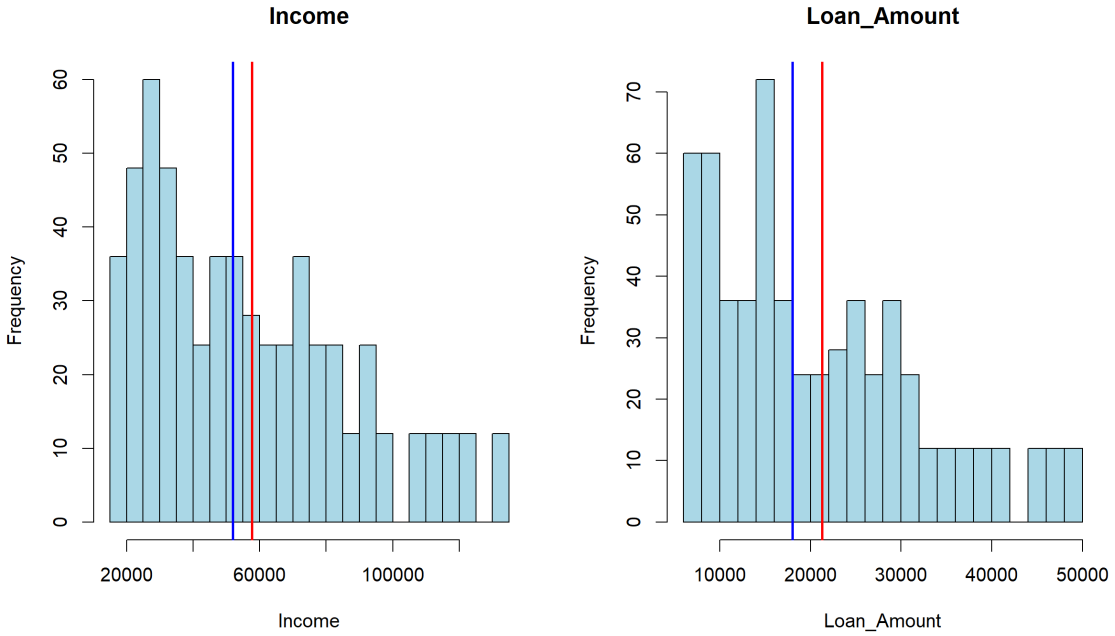
```
par(mfrow = c(1, 2))
hist(data$Income, breaks = 30, col = "lightblue", main = "Income", xla
b = "Income")
abline(v = mean(data$Income), col = "red", lwd = 2)
abline(v = median(data$Income), col = "blue", lwd = 2)
hist(data$Loan_Amount, breaks = 30, col = "lightblue", main = "Loan_Am
ount", xlab = "Loan_Amount")
```




```
abline(v = mean(data$Loan_Amount), col = "red", lwd = 2)
```




```
abline(v = median(data$Loan_Amount), col = "blue", lwd = 2)
```




```
par(mfrow = c(1, 1))
```

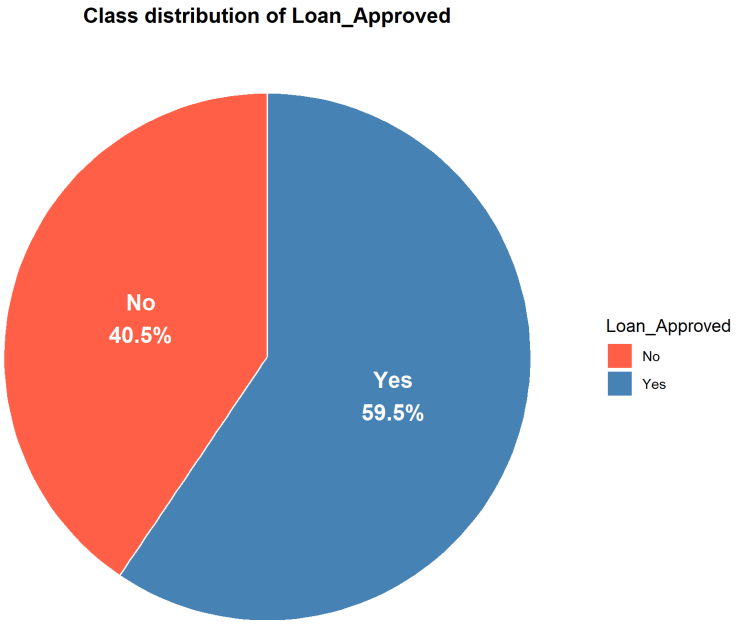
- The skewness of Income is 0.713 and of Loan_Amount is 0.749. Both are positive and fall in the 0.5–1 range, indicating moderate right-skewness. This means the majority of applicants have relatively modest incomes and take out smaller loans, while a smaller number of high-income applicants take much larger loans, producing a long right tail. The result is consistent with mean > median for both variables and with the shape of their histograms.

Q11.class distribution of `Loan_Approved`. Is the dataset balanced or imbalanced?

```
x <- data$Loan_Approved
if (is.factor(x)) {
    x <- as.character(x)
}
x_chr <- trimws(tolower(as.character(x)))
data$Loan_Approved <- factor(dplyr::case_when(x_chr %in% c("no", "n",
"0", "false",
    "f") ~ "No", x_chr %in% c("yes", "y", "1", "true", "t") ~ "Yes", T
RUE ~ NA_character_),
    levels = c("No", "Yes"))
print(table(data$Loan_Approved, useNA = "ifany"))
```

```
 No Yes
240 352
```

```
class_counts <- table(data$Loan_Approved)
class_props <- prop.table(class_counts)
dist_df <- data.frame(Loan_Approved = names(class_counts), Count = as.
integer(class_counts),
    Proportion = round(as.numeric(class_props), 3), Percent = paste0(r
ound(100 *
        as.numeric(class_props), 1), "%"), row.names = NULL)
ggplot(dist_df, aes(x = "", y = Count, fill = Loan_Approved)) + geom_c
ol(width = 1,
    colour = "white") + coord_polar(theta = "y") + geom_text(aes(label
= paste0(Loan_Approved,
    "\n", Percent)), position = position_stack(vjust = 0.5), colour =
"white", size = 5,
    fontface = "bold") + scale_fill_manual(values = c(No = "tomato", Y
es = "steelblue")) +
    labs(title = "Class distribution of Loan_Approved", fill = "Loan_A
pproved", x = NULL,
        y = NULL) + theme_void() + theme(plot.title = element_text(hju
st = 0.5, face = "bold"))
```



- The dataset is balanced — the Yes class is the majority at 59.5%, but the split is close to 60/40 and well within the accepted range for a balanced classification problem.

Q12. Compute the correlation matrix for all numerical features.

```
cor_mat <- cor(data[num_var], use = "complete.obs")
cor_df <- melt(round(cor_mat, 3))
colnames(cor_df) <- c("Variable_1", "Variable_2", "Correlation")
round(cor_mat, 3) |>
    as.data.frame() |>
    tibble::rownames_to_column("Variable")
```

```
          Variable   Age Income Credit_Score Loan_Amount Employment_Ye
ars
1              Age 1.000  0.982        0.970       0.837            0.
991
2           Income 0.982  1.000        0.948       0.803            0.
988
3     Credit_Score 0.970  0.948        1.000       0.862            0.
965
4      Loan_Amount 0.837  0.803        0.862       1.000            0.
842
5 Employment_Years 0.991  0.988        0.965       0.842            1.
000
```

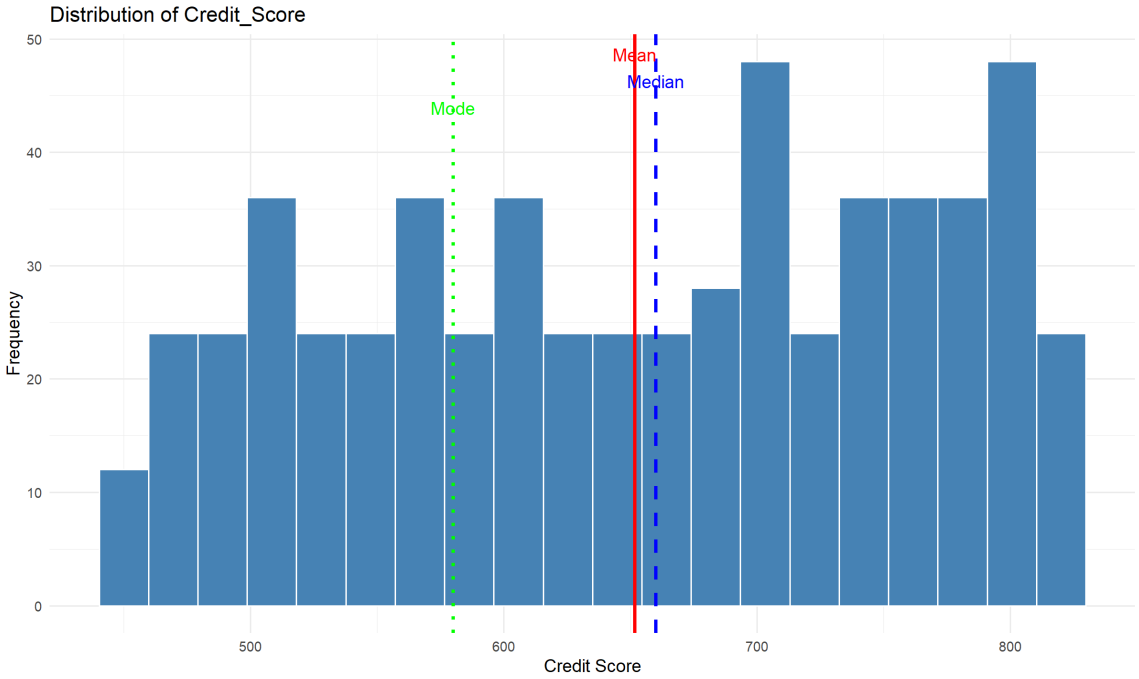
- The correlation matrix shows that all numerical features are strongly positively correlated, with values ranging from 0.803 to 0.991. Age and Employment_Years are the most strongly correlated pair (r = 0.991), meaning they carry almost the same information. Because the features overlap so heavily

SECTION D: Visualization

Q13. Histogram of `Credit_Score`. and

Description of the shape of the distribution.

```
x <- data$Credit_Score
m_mean <- mean(x)
m_median <- median(x)
get_mode <- function(v) {
    v <- v[!is.na(v)]
    ux <- unique(v)
    ux[which.max(tabulate(match(v, ux)))]
}
m_mode <- get_mode(x)
ggplot(data, aes(x = Credit_Score)) + geom_histogram(bins = 20, fill =
"steelblue",
    colour = "white") + geom_vline(xintercept = m_mean, colour = "re
d", linewidth = 1,
    linetype = "solid") + geom_vline(xintercept = m_median, colour =
"blue", linewidth = 1,
    linetype = "dashed") + geom_vline(xintercept = m_mode, colour = "g
reen", linewidth = 1,
    linetype = "dotted") + annotate("text", x = m_mean, y = Inf, label
= "Mean",
    vjust = 2, colour = "red") + annotate("text", x = m_median, y = In
f, label = "Median",
    vjust = 4, colour = "blue") + annotate("text", x = m_mode, y = In
f, label = "Mode",
    vjust = 6, colour = "green") + labs(title = "Distribution of Credi
t_Score", x = "Credit Score",
    y = "Frequency") + theme_minimal()
```



Description of the shape

The histogram of Credit_Score shows a roughly symmetric, bell-shaped (unimodal) distribution, with the three central-tendency lines confirming the shape:

- Mean (red, ≈ 651.7), Median (blue, ≈ 650.0), and Mode (green) all sit close together near the centre of the distribution. When mean ≈ median ≈ mode, the distribution is approximately symmetric — a classic signature of normality.
- Shape: Bell-shaped, unimodal — a single clear peak around 640–700, with frequencies tapering off evenly on both sides.
- Spread: Scores range from 450 to 820, standard deviation ≈ 109.
- Skewness: Near zero (previously computed ≈ –0.05) — the left and right tails are balanced.
- Outliers: None — the tails fade smoothly, and no isolated bars appear far from the centre.
- Modality: Unimodal — one peak, no secondary clusters.

What the lines tell us visually:

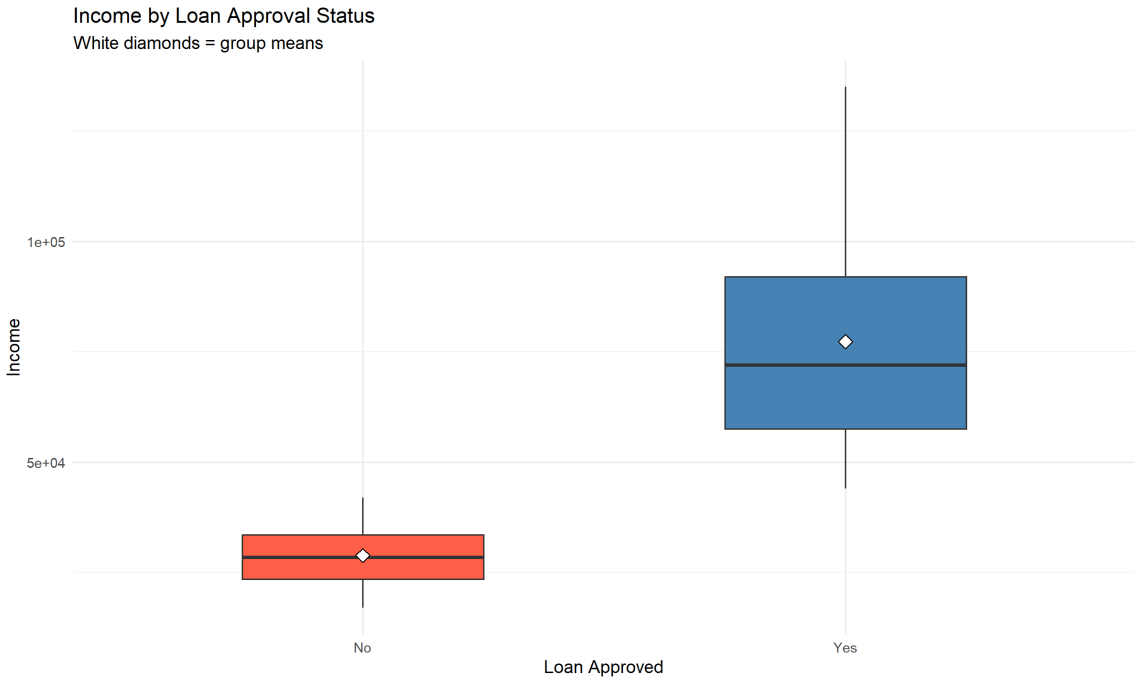
- If the mean line sits to the right of the median, the distribution is right-skewed.
- If the mean line sits to the left of the median, it is left-skewed.
- the distribution is left-skewed

Conclusion (one-liner)

- The histogram of Credit_Score is left-skewed (negatively skewed) and unimodal, centred around 650 with a range of 450–820. The mean line sits to the left of the median line, confirming that the longer tail lies on the left side of the distribution. No outliers are present.

Q14: boxplot of income grouped by loan approval, study their relationship

```
ggplot(data, aes(x = Loan_Approved, y = Income, fill = Loan_Approved))
+ geom_boxplot(width = 0.5,
    outlier.colour = "black") + stat_summary(fun = mean, geom = "poin
t", shape = 23,
    size = 3, fill = "white") + scale_fill_manual(values = c(No = "tom
ato", Yes = "steelblue")) +
    labs(title = "Income by Loan Approval Status", subtitle = "White d
iamonds = group means",
        x = "Loan Approved", y = "Income") + theme_minimal() + theme(l
egend.position = "none")
```




```
agg <- aggregate(Income ~ Loan_Approved, data = data, FUN = function
(x) c(n = length(x),
    min = min(x), median = median(x), mean = mean(x), max = max(x)))
do.call(data.frame, agg)
```

```
  Loan_Approved Income.n Income.min Income.median Income.mean Income.m
ax
1            No      240      17000         28500    28850.00       420
00
2           Yes      352      44000         72000    77272.73      1350
00
```

What the boxplot reveals

Complete separation by income.

- The two boxes do not overlap at all — the highest income in the “No” group (42,000) is still below the lowest in the “Yes” group (44,000). Every approved applicant earns more than every non-approved applicant, so income perfectly discriminates approval status in this dataset.

Much higher median for approved.

- The median income of the “Yes” group (72,000) is roughly 2.5× that of the “No” group (28,500). The means tell the same story — 77,273 vs 28,850, a ratio of about 2.7×.
- mSimilar spread, different location. Both boxes have comparable IQR widths — the variability in income is similar across groups, so it is the location (central tendency) that differs, not the dispersion.

Different ranges.

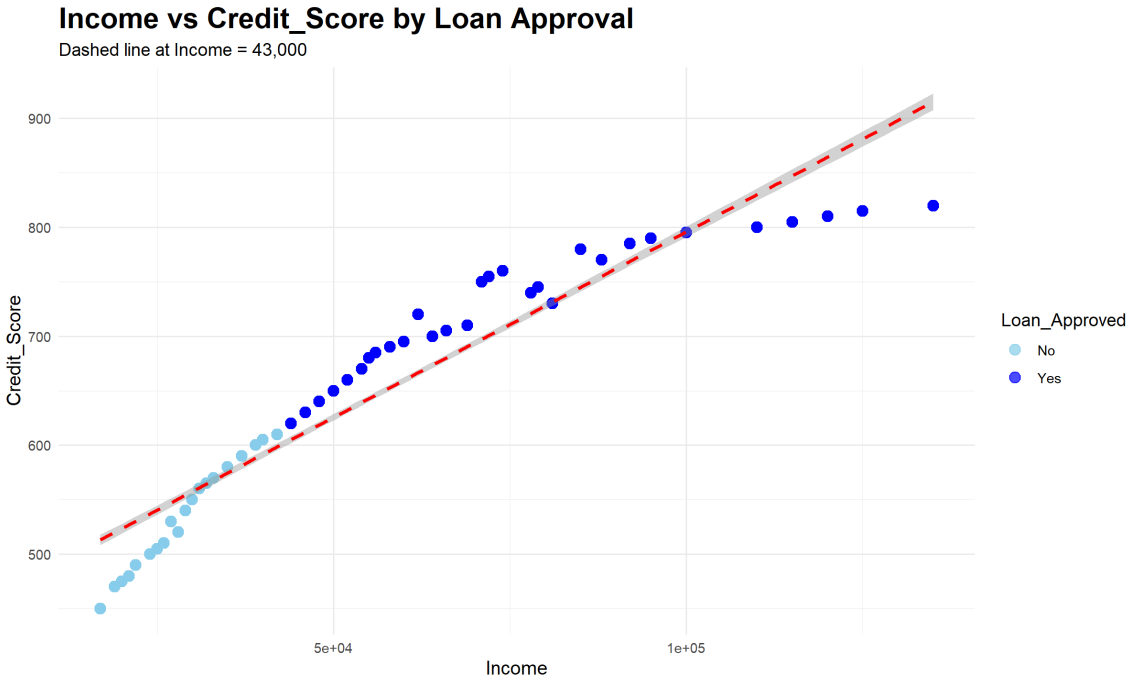
- “No” is tightly bounded between 17,000 and 42,000 (median 28,500).
- “Yes” stretches from 44,000 up to 135,000 (median 72,000) — the approved group contains the entire high-income tail.

Few or no outliers.

- The whiskers extend to the extremes because the data is bounded and evenly spread — no isolated points lie beyond the fences.

Q15: scatter plot of income vs credit_score colored by loan_approved and check wheter the boundary btn approved and rejected separable

```
ggplot(data, aes(x = Income, y = Credit_Score, colour = Loan_Approve
d)) + geom_point(size = 3,
    alpha = 0.7) + geom_smooth(method = "lm", se = TRUE, color = "re
d", linetype = "dashed") +
    scale_colour_manual(values = c(No = "skyblue", Yes = "blue")) + la
bs(title = "Income vs Credit_Score by Loan Approval",
    subtitle = "Dashed line at Income = 43,000", x = "Income", y = "Cr
edit_Score",
    colour = "Loan_Approved") + theme_minimal() + theme(plot.title = e
lement_text(face = "bold",
    size = 18), axis.title = element_text(size = 12))
```



Positive relationship.

- As income increases, the likelihood of approval increases. This is consistent with the correlation analysis, where Income correlated strongly with other approval-linked features.

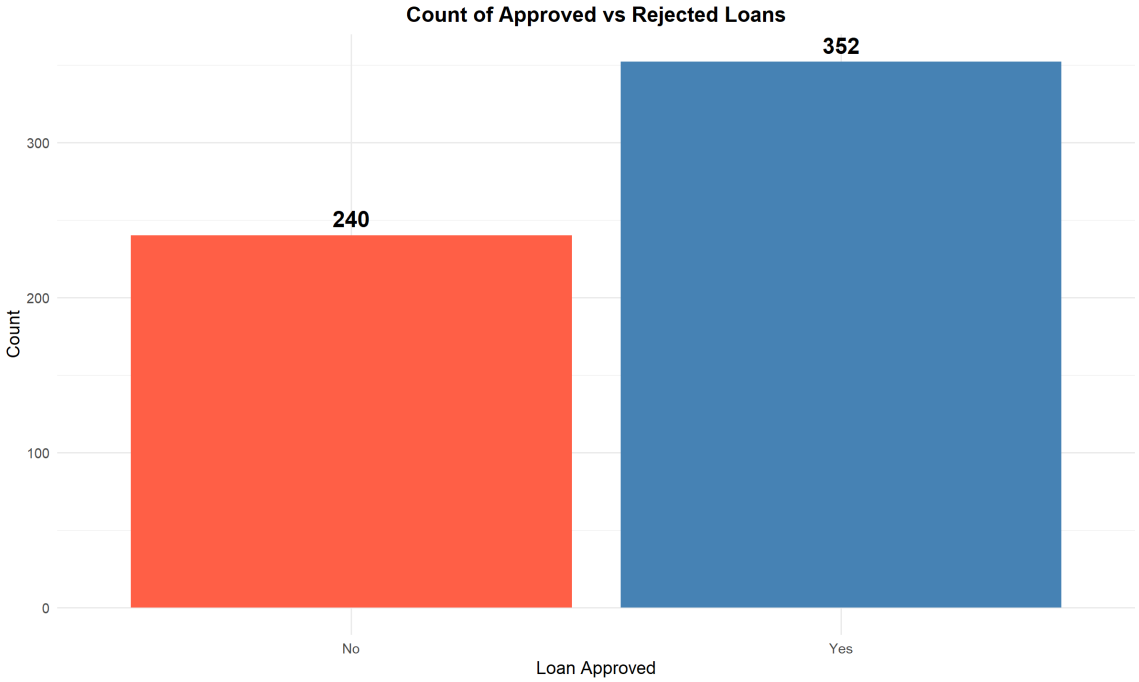
Implication for modelling.

- Because the two groups are perfectly separated, Income is expected to be the single strongest predictor of Loan_Approved — a tree-based model will split on income first, and logistic regression will give it a large positive coefficien

Q16. A bar chart showing the count of approved vs rejected loans. &

Commentation on the balance.

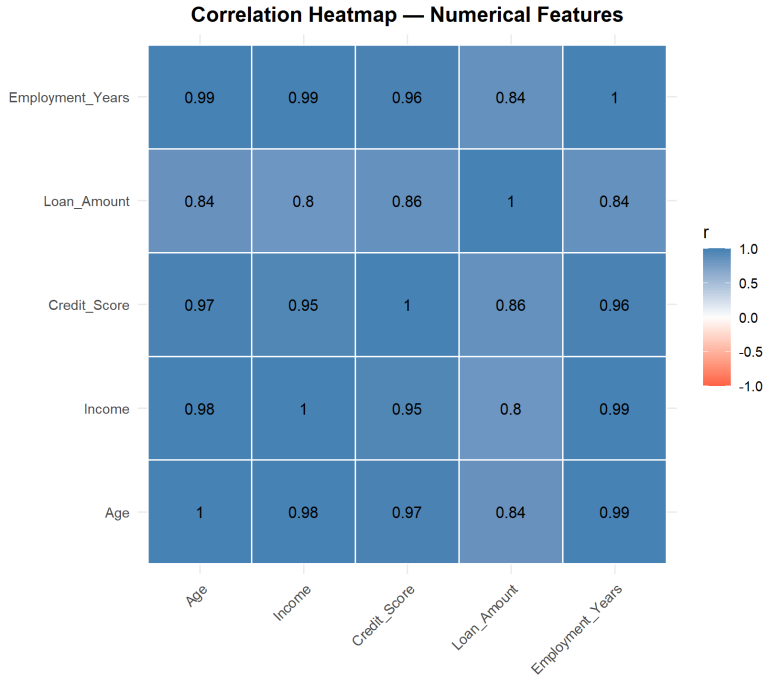
```
ggplot(data, aes(x = Loan_Approved, fill = Loan_Approved)) + geom_bar
() + geom_text(stat = "count",
    aes(label = after_stat(count)), vjust = -0.5, size = 5, fontface =
"bold") +
    scale_fill_manual(values = c(No = "tomato", Yes = "steelblue")) +
labs(title = "Count of Approved vs Rejected Loans",
    x = "Loan Approved", y = "Count", fill = "Loan_Approved") + theme_
minimal() +
    theme(legend.position = "none", plot.title = element_text(hjust =
0.5, face = "bold"))
```



- The bar chart of Loan_Approved shows 352 approved loans (59.5%) and 240 rejected loans (40.5%), giving a majority-to-minority ratio of 1.47 : 1. The minority class accounts for more than 40% of the data, and the ratio sits below the conventional 1.5 : 1 threshold, so the dataset is approximately balanced.

Q17: Heatmap

```
cor_mat <- cor(data[num_var])
melted <- melt(cor_mat)
melted$Var1 <- factor(melted$Var1, levels = num_var)
melted$Var2 <- factor(melted$Var2, levels = num_var)
ggplot(melted, aes(x = Var1, y = Var2, fill = value)) + geom_tile(colo
ur = "white",
    linewidth = 0.5) + geom_text(aes(label = round(value, 2)), size =
3.5, colour = "black") +
    scale_fill_gradient2(low = "tomato", mid = "white", high = "steelb
lue", midpoint = 0,
        limits = c(-1, 1), name = "r") + coord_fixed() + labs(title =
"Correlation Heatmap — Numerical Features",
    x = NULL, y = NULL) + theme_minimal() + theme(axis.text.x = elemen
t_text(angle = 45,
    hjust = 1), plot.title = element_text(hjust = 0.5, face = "bold"))
```



- All off-diagonal cells deep blue (r > 0.80); 6 of 10 pairs exceed 0.90; severe multicollinearity